

Eric Gan

510-358-6280 | ericgan64@gmail.com | [linkedin.com/in/ericgan64](https://www.linkedin.com/in/ericgan64) | eri-gon.github.io/EricGan/

EDUCATION

UC San Diego - 3.87 GPA

Bachelor of Arts in Data Science

La Jolla, CA

Sep. 2023 – Jun 2027

EXPERIENCE

Undergraduate Research Assistant

Knight Lab, UC San Diego

Feb. 2025 – June 2025

San Diego, CA

- Performed data scraping and processing of large-scale microbiome datasets using Python and Bash
- Conducted QIIME 2 analyses to study microbial community structure and diversity
- Developed and optimized workflows to streamline microbiome data cleaning and visualization

Food Service Assistant Manager

Philmont Scout Ranch

June 2024 – August 2024, June 2025 – August 2025

Cimarron, NM

- Supervised food service operations serving up to 300 people per meal
- Managed inventory, ordering, and logistics across multiple facilities
- Created SOPs to improve operational efficiency and consistency
- Trained and led staff to ensure safe and efficient service

Co-Founder

333Jugglers, UC San Diego

December 2024 – Present

San Diego, CA

- Founded and scaled UCSD juggling club from concept to official recognition
- Led recruitment, events, and weekly skill workshops
- Managed logistics, partnerships, and long-term sustainability planning

PROJECTS

1D Convolution Visualization Playground | *React, JavaScript, Web Audio API, D3* | [Live Demo](#)

Mar 2026

- Built an interactive web application to visualize discrete convolution with step-by-step kernel sliding
- Designed interactive controls for signals and kernels to improve intuition for linear systems

SoCalGuessr Image Classification | *PyTorch, MobileNetV3, Computer Vision*

Mar 2026

- Built and evaluated a 6-class image classification model using MobileNetV3-Small
- Applied transfer learning by modifying final layers for custom classification
- Achieved 84% validation accuracy over 20 epochs
- Optimized lightweight architecture (1.5M parameters) for efficiency

Gait Pattern Analyzer | *Python, D3, JavaScript, HTML, CSS*

June 2025

- Developed interactive visualizations to analyze gait data from neurological datasets
- Used Python for data processing and D3.js for dynamic comparisons
- Designed UI for exploring patient-level gait differences

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React, PyTorch, scikit-learn, pandas, NumPy, Matplotlib, D3.js

Tools & Technologies: Git, Bash, Web Audio API, QIIME 2, VS Code, IntelliJ, Cursor, Antigravity